

UNIT OPERATIONS

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Introduction

- ❖ The history of the development of pharmaceutical technology clearly shows that before many years back, the manufacture of pharmaceuticals is in ***small scale level.***
- ❖ But, nowadays, due to many convincing reasons, these is changed to large scale manufacture:
 - **Economic reasons**
 - **Accuracy of production**
 - **Complexity of modern therapy**
- ❖ Although the end product of the manufacturing processes differs considerably, the stages involved shows similarities.



Introduction...

- ❖ For example: mixing is part of manufacturing process for products in
 - Purification of a substance by crystallization,
 - Manufacture of a mixed powders
 - Compressed tablets etc.



Introduction...

❖ **Table:** Unit operations involved in different manufacturing process

Manufacturing process	Unit operations involved
Purification of substance by crystallization	• Mixing • Filtration • Evaporation • Centrifugation
Manufacture of mixed powders	• Size reduction • Size separation • Mixing
Manufacture of compressed tablets	• Size reduction • Size separation • Mixing • Drying • Compaction



Introduction...

- ❖ Therefore, the main unit operations in pharmaceutical technology include:
 - Evaporation
 - Extraction
 - Size reduction
 - Size separation
 - Distillation
 - Mixing
 - Crystallization
 - Centrifugation
 - Drying
 - Filtration
- Hence, these individual operations that make up the whole process are called *unit operations*.



Fundamentals of Unit Operations

A. Fluid flow

- ❖ Many raw materials and finished pharmaceuticals are in the form of fluid. It may be liquid, gas or fluidized powders and granules
- ❖ These have to be transported and processed. Therefore characteristics of fluid flow will affect the production process
- ❖ The flow of fluids can be best visualized by reference to a classical experiment (**Reynold's experiment**)

Fundamentals...

Reynold's experiment

- ❖ Into a flow through a glass tube (Fig.), he injected a dye to observe the nature of flow.

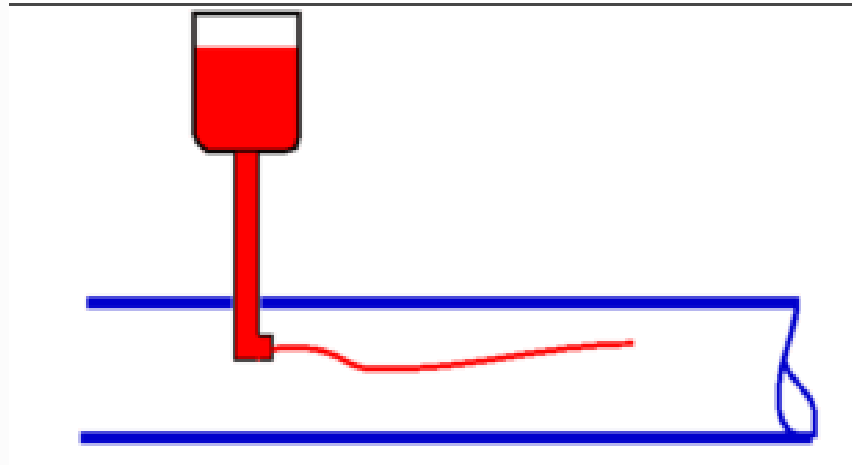


Figure: Reynolds Experiment

Fundamentals...

Observations:

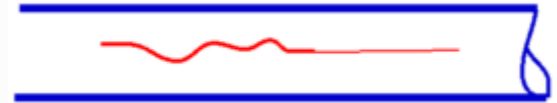
❖ At low speed:

- Colored thread remains undistributed in the center of water and moves steadily along the tube without mixing
- **Streamline flow or laminar flow**



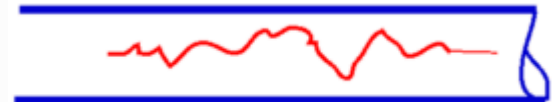
❖ At moderate speed

- A point is reached (critical speed) at which the thread begins to waver although no mixing occurs
- **Transitional flow**

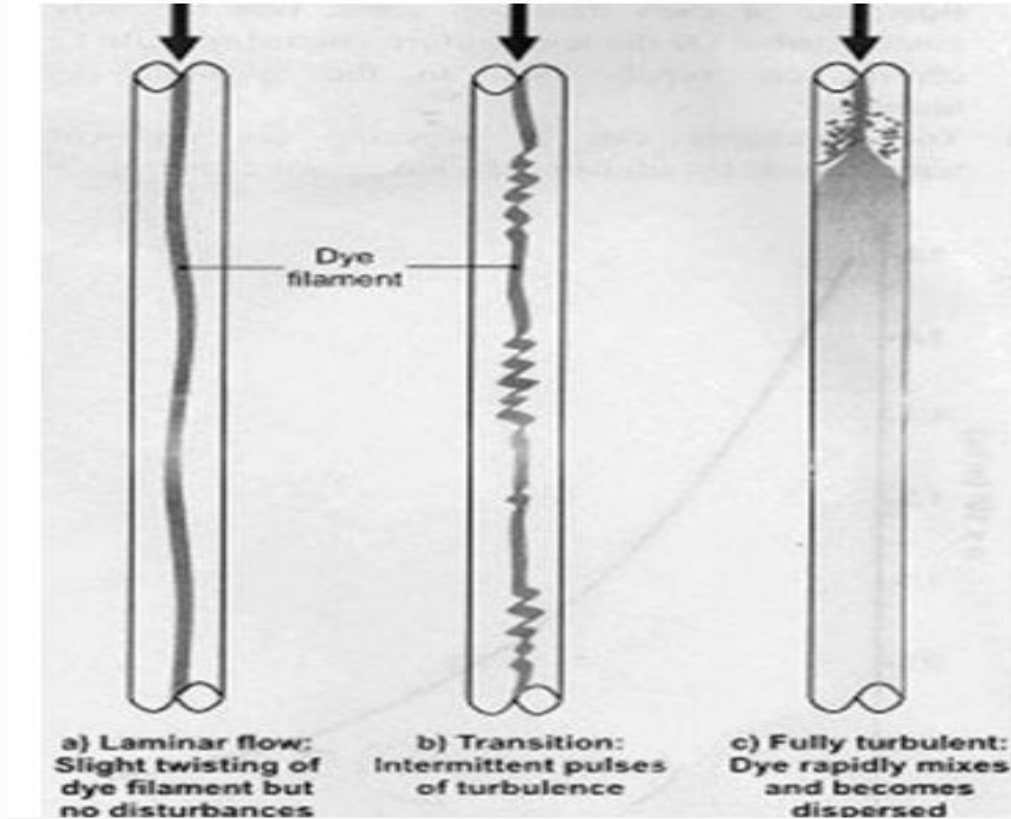


❖ At high speed

- Eddies occur and colored liquid mixes with the bulk of water
- **Turbulent flow**



Fundamentals...





Fundamentals...

The significance of Reynold's number, Re

$$Re = \frac{\rho v d}{\mu}$$

Re – Reynold's number

ρ – density

v – velocity of fluid

d – diameter

μ – viscosity

- ❖ The Reynold's number can be used to determine the type of flow at particular circumstances
 - ✓ **If $Re < 2000$ streamline flow**
 - ✓ **If $Re > 4000$ turbulent flow**



Fundamentals...

- ❖ In Reynold's experiment, progressive increase in speed cause a change from laminar to turbulent flow
- ❖ These change would also have occurred if the diameter of the tube was increased while maintaining the velocity, or if the fluid was replaced by one of higher density
- ❖ On the other hand, an increase viscosity could promote a change in the opposite direction
- ❖ From the formula we can identify the factors which will affect the flow. These can include:
 - *Diameter of the pipe*
 - *Velocity of the fluid*
 - *Density of the fluid*
 - *Viscosity of the fluid*



Fundamentals...

B. Heat transfer

- ❖ Heat is a form of energy which is transferred from a region of higher temperature to a region of lower temperature
- ❖ Heat has a lot of pharmaceutical applications, of these:
 - Melting of materials
 - Hasten dissolution process
 - Sterilization
 - Evaporation
 - Drying of granules for tablet production
 - Heating of air to facilitate coating of tablets



Fundamentals...

❖ Heat is transferred by three mechanisms:

- Conduction
- Convection
- Radiation

1. Conduction

- ❖ It is the most widely understood mechanism of heat transfer
- ❖ It is the main method by which heat is transferred through solids

E.g. heating of the whole rod when one end is heated



Fundamentals...

2. Convection

- ❖ Here the transfer of heat takes place from one point to another in a body of fluid by actual movement of particles

E.g. heating of water in glass beaker

3. Radiation

- ❖ Heat transmission by radiation occurs by energy transfer through space by electromagnetic radiation

E.g. heat from sun



Fundamentals...

Factors affecting heat transfer

❖ Area

- Heating should take place as large SA as possible

❖ T° gradient:-

- It should be as suitable and as great as possible

❖ Materials of construction

- The plant should be made from materials of suitable thermal conductivity

❖ General design

- Cleanliness
- Condensate removal



Fundamentals...

C. Mass transfer

- ❖ It is the process that involves molecular and convective transport of atoms and molecules within physical systems
- ❖ **Examples:**
 - Evaporation of water
 - Extraction of drug
 - Drying of pharmaceuticals
- ❖ There are two types of mass transfer
 - ***Solid/fluid mass transfer***
 - ***Fluid/fluid mass transfer***



Fundamentals...

I. Solid/fluid mass transfer

- ❖ Materials immersed in a solvent contains two layers (a stationary boundary layer and bulk of the fluid)
- ❖ The transport of the molecules (mass transfer) takes place in two stages:
 - ✓ **Molecular diffusion:-** movement of molecules through the boundary layers
 - ✓ **Eddy diffusion:-** once the material has passed through the boundary layer, mass transfer takes place by the bulk movement of the solution

II. Fluid/fluid mass transfer

- ❖ Here mass transfer takes place between two immiscible fluids



Fundamentals...

Factors affecting the rate of mass transfer

- ❖ *Surface area*
- ❖ *Film thickness/boundary layer*
- ❖ *Concentration gradient*
- ❖ The rate of mass transfer can be increased by:
 - ✓ **Agitation**
 - ✓ **Elevated T°**
 - ✓ **Size reduction**